



Glioma Grading Clinical and Mutation Features

Donated on 12/13/2022

Gliomas are the most common primary tumors of the brain. They can be graded as LGG (Lower-Grade Glioma) or GBM (Glioblastoma Multiforme) depending on the histological/imaging criteria. Clinical and molecular/mutation factors are also very crucial for the grading process. Molecular tests are expensive to help accurately diagnose glioma patients. In this dataset, the most frequently mutated 20 genes and 3 clinical features are considered from TCGA-LGG and TCGA-GBM brain glioma projects. The prediction task is to determine whether a patient is LGG or GBM with a given clinical and molecular/mutation features. The main objective is to find the optimal subset of mutation genes and clinical features for the glioma grading process to improve performance and reduce costs.

Dataset Characteristics

Tabular, Multivariate

Feature Type

Real, Categorical, Integer

Subject Area

Health and Medicine

Instances

839

Associated Tasks

Classification, Other

Features

23

Dataset Information

For what purpose was the dataset created?



DOWNLOAD (301.8 KB)



IMPORT IN PYTHON

CITE

” 1 citations

👁 27267 views

Citations/Acknowledgement

If you use this dataset, please cite:

Tasci, E., Zhuge, Y., Kaur, H., Camphausen, K., & Krauze, A. V. (2022). Hierarchical Voting-Based Feature Selection and Ensemble Learning Model Scheme for ...



Keywords

Mutation

Tumor grading

Gliomas are the most common primary tumors of the brain. They can be graded as LGG (Lower-Grade Glioma) or GBM (Glioblastoma Multiforme) depending on the histological/imaging criteria. Clinical and molecular/mutation factors are also very crucial for the grading process. Molecular tests are expensive t...

SHOW MORE ▾

Who funded the creation of the dataset?

The Cancer Genome Atlas (TCGA) Project – NCI

What do the instances in this dataset represent?

In this dataset, the instances represent the records of patients who have brain glioma. The dataset was constructed based on TCGA-LGG and TCGA-GBM brain glioma projects.

...

SHOW MORE ▾

Are there recommended data splits?

No. We suggest 10-fold cross-validation for feature selection, classification etc.

Does the dataset contain data that might be considered sensitive in any way?

There is information about race, age, and gender of the patient.

Was there any data preprocessing performed?

Yes.

The original and preprocessed files differ in the following ways:

- There are 23 instances in the original file where Gender, Age_at_diagnosis, or Race feature values are '--', or 'not reported'. These instances were filtered out in the preprocessed dataset.
- Despite being present in the original dataset, we do not include the columns Project, Case_ID, and Primary_Diagnosis columns in the preprocessed dataset.
- Age_at_diagnosis feature values were converted from string to continuous value by adding day information to the corresponding year information in the dataset as a floating-point number for the preprocessing stage.

Creators

Erdal Tasci

erdal.tasci@nih.gov



Radiation Oncology Branch (ROB),
National Cancer Institute (NCI),
National Institutes of Health (NIH),
Building 10

Kevin Camphausen

camphauk@mail.nih.gov



Radiation Oncology Branch (ROB),
National Cancer Institute (NCI),
National Institutes of Health (NIH),
Building 10

Andra Valentina Krauze

andra.krauze@nih.gov



Radiation Oncology Branch (ROB),
National Cancer Institute (NCI),
National Institutes of Health (NIH),
Building 10

Ying Zhuge

zhugey@mail.nih.gov



Radiation Oncology Branch (ROB),
National Cancer Institute (NCI),
National Institutes of Health (NIH),
Building 10

DOI

[10.24432/C5R62J](https://doi.org/10.24432/C5R62J)

License

All processed and unprocessed files also exist in this directory.

Below is a list of the additional columns of the original dataset file (and their corresponding description):

- Project column represents corresponding TCGA-LGG or TCGA-GBM project names.
- Case_ID column refers to the related project Case_ID information.
- Primary_Diagnosis column provides information related to the type of primary diagnosis.

SHOW LESS ^

Has Missing Values?

No

This dataset is licensed under a [Creative Commons Attribution 4.0 International](#) (CC BY 4.0) license.

This allows for the sharing and adaptation of the datasets for any purpose, provided that the appropriate credit is given.

Introductory Paper ^

[Hierarchical Voting-Based Feature Selection and Ensemble Learning Model Scheme for Glioma Grading with Clinical and Molecular Characteristics](#)

By E. Tasci, Y. Zhuge, Harpreet Kaur, K. Camphausen, A. Krauze. 2022

Published in International Journal of Molecular Sciences

Variables Table ^

Variable Name	Role	Type	Demographic	Description	Units	Missing Valu
Grade	Target	Categorical		Glioma grade class information (0 = "LGG"; 1 = "GBM")	N/A	no

Variable Name	Role	Type	Demographic	Description	Units	Missing Valu
Gender	Feature	Categorical	Gender	Gender (0 = "male"; 1 = "female")	N/A	no
Age_at_diagnosis	Feature	Continuous	Age	Age at diagnosis with the calculated number of days	years	no
Race	Feature	Categorical	Race	Race (0 = "white"; 1 = "black or african American"; 2 = "asian"; 3 = "american indian or alaska native")	N/A	no
IDH1	Feature	Categorical		isocitrate dehydrogenase (NADP(+))1 (0 = NOT_MUTATED; 1 = MUTATED)	N/A	no
TP53	Feature	Categorical		tumor protein p53 (0 = NOT_MUTATED; 1 = MUTATED)	N/A	no
ATRX	Feature	Categorical		ATRX chromatin remodeler (0 = NOT_MUTATED; 1 = MUTATED)	N/A	no
PTEN	Feature	Categorical		phosphatase and tensin homolog (0 =	N/A	no

Variable Name	Role	Type	Demographic	Description	Units	Missing Value
				NOT_MUTATED; 1 = MUTATED)		
EGFR	Feature	Categorical		epidermal growth factor receptor (0 = NOT_MUTATED; 1 = MUTATED)	N/A	no
CIC	Feature	Categorical		capicua transcriptional repressor (0 = NOT_MUTATED; 1 = MUTATED)	N/A	no
MUC16	Feature	Categorical		mucin 16, cell surface associated (0 = NOT_MUTATED; 1 = MUTATED)	N/A	no
PIK3CA	Feature	Categorical		phosphatidylinositol-4,5-bisphosphate 3-kinase catalytic subunit alpha (0 = NOT_MUTATED; 1 = MUTATED)	N/A	no
NF1	Feature	Categorical		neurofibromin 1 (0 = NOT_MUTATED; 1 = MUTATED)	N/A	no
PIK3R1	Feature	Categorical		phosphoinositide-3-kinase regulatory	N/A	no

Variable Name	Role	Type	Demographic	Description	Units	Missing Valu
				subunit 1 (0 = NOT_MUTATED; 1 = MUTATED)		
FUBP1	Feature	Categorical		far upstream element binding protein 1 (0 = NOT_MUTATED; 1 = MUTATED)	N/A	no
RB1	Feature	Categorical		RB transcriptional corepressor 1 (0 = NOT_MUTATED; 1 = MUTATED)	N/A	no
NOTCH1	Feature	Categorical		notch receptor 1 (0 = NOT_MUTATED; 1 = MUTATED)	N/A	no
BCOR	Feature	Categorical		BCL6 corepressor (0 = NOT_MUTATED; 1 = MUTATED)	N/A	no
CSMD3	Feature	Categorical		CUB and Sushi multiple domains 3 (0 = NOT_MUTATED; 1 = MUTATED)	N/A	no
SMARCA4	Feature	Categorical		SWI/SNF related, matrix associated, actin dependent regulator of	N/A	no

Variable Name	Role	Type	Demographic	Description	Units	Missing Valu
				chromatin, subfamily a, member 4 (0 = NOT_MUTATED; 1 = MUTATED)		
GRIN2A	Feature	Categorical		glutamate ionotropic receptor NMDA type subunit 2A (0 = NOT_MUTATED; 1 = MUTATED)	N/A	no
IDH2	Feature	Categorical		isocitrate dehydrogenase (NADP(+)) 2 (0 = NOT_MUTATED; 1 = MUTATED)	N/A	no
FAT4	Feature	Categorical		FAT atypical cadherin 4 (0 = NOT_MUTATED; 1 = MUTATED)	N/A	no
PDGFRA	Feature	Categorical		platelet-derived growth factor receptor alpha (0 = NOT_MUTATED; 1 = MUTATED)	N/A	no

Dataset Files



File	Size
TCGA_GBM_LGG_Mutations_all.csv	258.8 KB
TCGA_InfoWithGrade.csv	42.7 KB

Reviews



There are no reviews for this dataset yet.

[LOGIN TO WRITE A REVIEW](#)

UC Irvine
Machine Learning
Repository

THE PROJECT

[About Us](#)[CML](#)[National Science Foundation](#)

NAVIGATION

[Home](#)[View Datasets](#)[Donate a Dataset](#)

LOGISTICS

[Contact](#)[Privacy Notice](#)[Feature Request or Bug Report](#)